

Application Protocol Data Unit (APDU)

An APDU is a communication unit between a smart card reader and a smart card



Why was APDU created?

Why does Ledger use it?

How old is it?

What other familiar things is it used for?

How wide is the scope of usage?

Are ledger devices considered smart cards?

Do APDUs refer to ip addresses?

Does it use TCP?

Protocol Data Unit (PDU)

A single unit of information transmitted among peer entities of a computer network



Examples:

- TCP: segment
- UDP: datagram
- Internet Layer: packet
- Link Layer: frame
- Physical Layer: bit (more generally symbol)
- ATM: cell
- APDU

Service Data Unit (SDU)

...

Smart Card

A smart card (SC), chip card, or integrated circuit card (ICC or IC card), is a card used to control access to a resource.



Examples

- SIM card

1967: Helmut Gröttrup filed a patent a tamper proof identification switch based on a semiconductor device with contactless communication via inductive coupling

1970: Kunitaka Arimura filed a patent for incorporating an integrated circuit into a plastic card

1974: Roland Moreno patented a secured memory card later dubbed the "smart card"

1983: Telephone cards

1991: First SIM card manufactured

1992: French Debit Card

APDU Structure

Command APDU

Field Name	Bytes	Description
CLA	1	Instruction class - type of command
INS	1	Instruction Code - specific command
P1	1	Parameter 1
P2	1	Parameter 2
L _c	0 1 3	Specify the length of data N _c
Command Data	N _c	
L _e	0 1 2 3	Encodes maximum response length

Response APDU

Field Name	Bytes	Description
Response Data	< N _e	Response Data
Status	2	Success, failure ???

Bitcoin APDU

Commands are executed with a single APDU, as opposed to being broken up into multiple

What does the get version APDU look like?

What are the different CLAs used for?

Are the different CLAs for different currencies, like ethereum?

Is there a CLA for bitcoin cash?

CLA	INS	Name
E1	00	GET_EXTENDED_PUBKEY
E1	01	REGISTER_WALLET
E1	02	GET_WALLET_ADDRESS
E1	03	SIGN_PSBT
E1	04	GET_MASTER_FINGERPRINT
E1	10	SIGN_MESSAGE